Appendix

A1. Data Constraints and the Scope of Minor-Party Analysis

Table 3 lists minor parties excluded from the empirical analysis because they lack legislative representation, are not covered by party-specific legislative performance items in the Taiwan Election and Democratization Study (TEDS) or have too few party-list respondents to support reliable statistical inference. Parties without Legislative Yuan seats are often omitted from performance evaluation batteries, and electorally marginal parties typically generate extremely small cell sizes in the survey. Including these cases would introduce substantial noise and undermine the stability of the estimates. The analysis therefore focuses on minor parties that either surpassed the 5 percent party-list threshold or were sufficiently represented and measured in the TEDS.

Table 3. Minor Parties Excluded from the Analysis Due to Lack of Legislative Representation, Questionnaire Coverage, or Insufficient Sample Size

Survey Year	Party	Party list vote count / N	Member in Legislative Yuan	Legislative Party Performance Questionnaire
2016TEDS	TSU	27 / 1248	0	X
	GPT	41 / 1248	0	X
2020TEDS	GPT	43 / 1347	0	X
	PFP	46 / 1347	0	X

Note: The total sample sizes reported have been adjusted after removing missing values.

A2. From Correlation to Causation: Evaluating Legislative Satisfaction Under Matched Designs

Figure 6 compares the estimated effects of retrospective satisfaction with legislative performance in the preceding four-year term on party-list voting using two approaches: a standard multinomial logit model (red) and models estimated after matching on observed confounders (blue). The comparison demonstrates that the relationship between satisfaction and voting persists once individuals are balanced on partisan identity, ideology, prior vote choice, demographics, and nomination context, thereby supporting the interpretation that retrospective legislative evaluations meaningfully shape voter decisions rather than simply reflecting prior preferences. This approach follows best practice in observational causal inference, whereby matching is used as a nonparametric preprocessing step to approximate balanced treatment groups before model estimation (Ho, Imai, King, & Stuart 2007; Rosenbaum & Rubin 1983; Morgan & Winship 2015: 110).

Across election years and parties, the matched estimates typically remain close to the conventional regression coefficients in both sign and magnitude, indicating that the observed performance effects are not artifacts of functional form or driven solely by pre-existing partisan attachments. In several cases—including the NPP in 2016 and 2020 and the TPP in 2024—the matched effects are even slightly stronger, suggesting

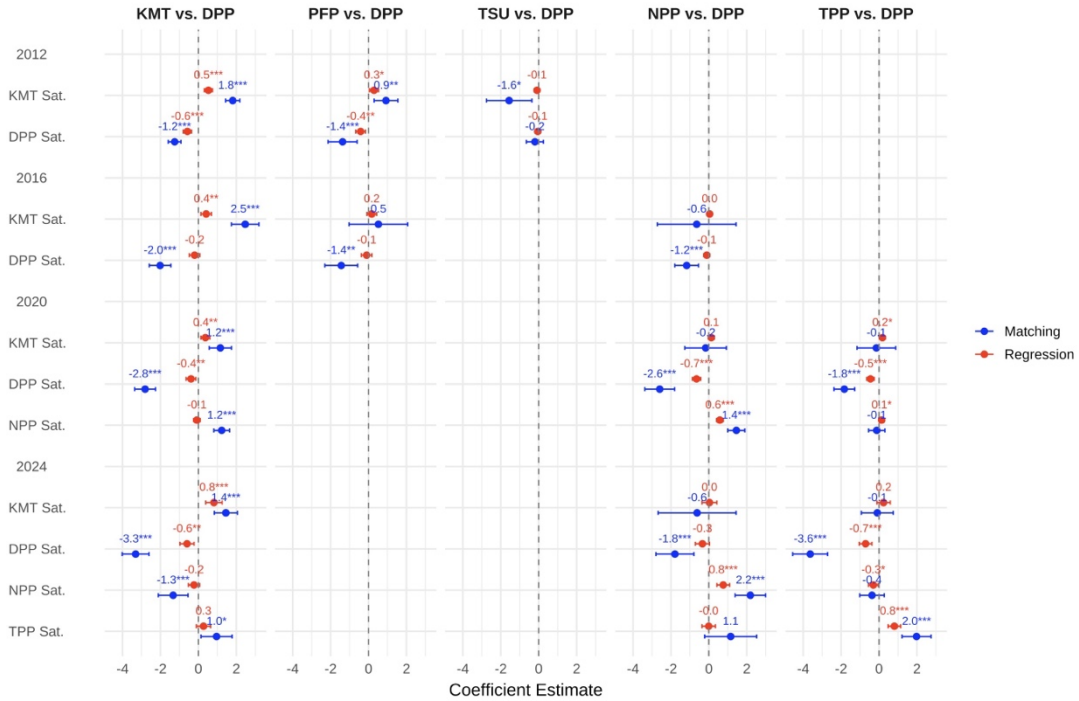
that when comparisons are restricted to voters with comparable political predispositions, differences in party-list choices are more clearly structured by perceptions of legislative competence and credibility. This pattern is consistent with the expectation that reducing imbalance sharpens substantive effects when causal relationships are genuine rather than modeling artifacts (Imbens & Rubin 2015; Stuart 2010).

Equally important, the matched models reduce spurious associations. Where regression-based coefficients appear inflated, matching constrains estimates to regions of common support and removes cases lacking plausible counterfactuals, thereby minimizing bias from extrapolation (Ho et al. 2007; Stuart 2010). This leads to more conservative but still statistically meaningful effects, reinforcing that our conclusions do not rely on unstable model assumptions or unbalanced samples. The stability of patterns across years further suggests that voters update their party-list choices in response to performance evaluations in systematic and theoretically consistent ways, aligning with research on retrospective accountability under mixed-member systems.

Taken together, the results align with the causal structure laid out in the DAG: once backdoor paths through partisan predispositions, ideology, prior vote, demographics, and nomination environment are blocked, satisfaction maintains a robust and substantively important association with minor-party support. While the design cannot eliminate all unobserved confounding, the persistence and consistency of matched

estimates indicate that retrospective legislative evaluations exert an independent influence on party-list voting under Taiwan's mixed-member system, consistent with best-practice claims of "causal leverage" in observational inference (Morgan & Winship 2015; Imbens & Rubin 2015).

Figure 6. Standard vs. Matched Multinomial Logit Estimates of Legislative Satisfaction on Party-List Vote Choice



Note: The figure compares coefficients from standard multinomial logit models (red) with models estimated after matching on observed confounders (blue). By restricting analysis to voters with comparable partisan identification, ideology, cocurrent vote behavior, demographics, and year difference, the matched design reduces extrapolation bias and blocks major backdoor confounding paths.

A3. Covariate Balance Diagnostics for Inverse Probability Weighting (IPW)

Figure 7 reports standardized mean differences (SMDs) for all pretreatment covariates before and after inverse probability weighting (IPW), separately for each election year (2012, 2016, 2020, and 2024). The figure demonstrates that, prior to weighting, substantial imbalance exists between respondents with high and low legislative satisfaction across multiple dimensions, including partisan attachments, ideological orientations, prior vote behavior, and demographic characteristics. Such imbalances indicate the presence of backdoor paths linking satisfaction to party-list vote choice through pre-existing political predispositions.

After IPW matching, covariate imbalance is substantially reduced across all election years. In each panel, post-matching SMDs fall within commonly accepted thresholds ($|SMD| < 0.1$), indicating that treated and control groups are statistically comparable on observed pretreatment attributes. This balance implies that the principal confounding paths identified in the causal DAG—specifically those operating through party identification, ideology, prior vote, and demographic characteristics—are effectively blocked.

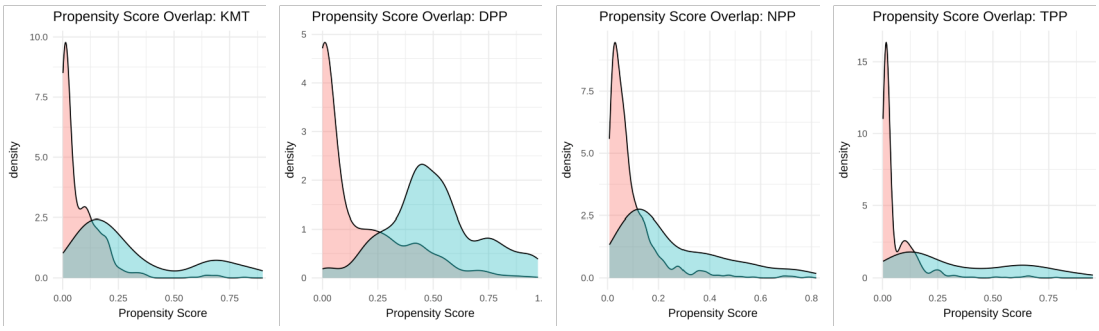
By enforcing common support and reweighting observations to achieve balance, IPW matching transforms the comparison from one between fundamentally different

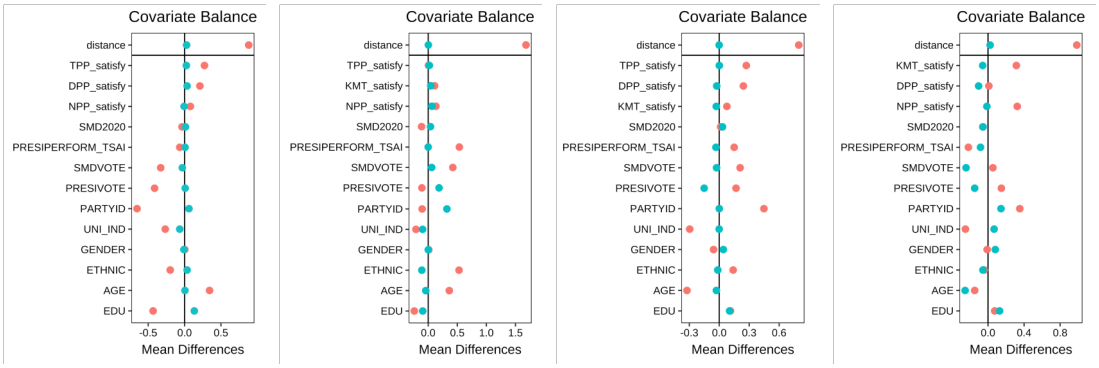
types of voters to one between otherwise similar individuals who differ primarily in their evaluations of legislative performance. This blocking function is crucial: it ensures that the estimated association between legislative satisfaction and party-list voting is not driven by underlying partisan or ideological differences that precede both attitudes and vote choice.

While matching cannot eliminate all unobserved confounding, the systematic reduction of imbalance across all observed covariates strengthens the credibility of the causal interpretation in the main analysis. Together with the stability of matched estimates across election years, the balance diagnostics in Figure 6 provide direct evidence that the observed performance effects are not artifacts of covariate imbalance, but instead reflect differences that persist after conditioning on a comprehensive set of pretreatment characteristics.

Figure 7. IPW Covariate Balance Diagnostics, 2012–2024

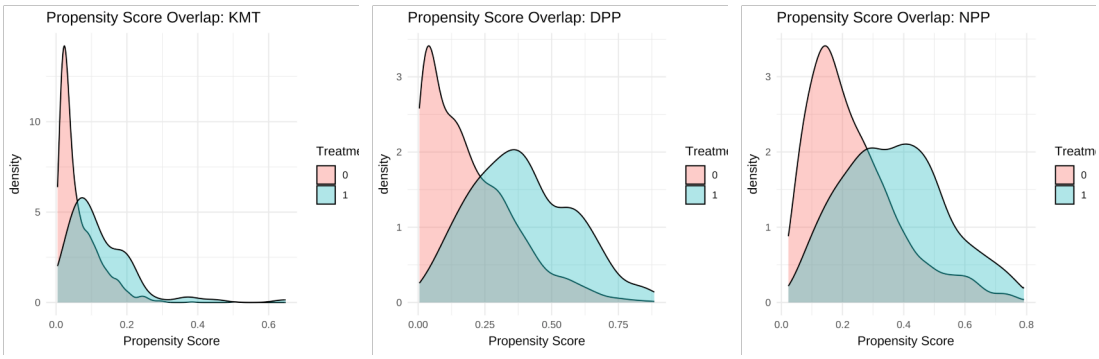
2024			
KMT Sat.	DPP Sat.	NPP Sat.	TPP Sat.

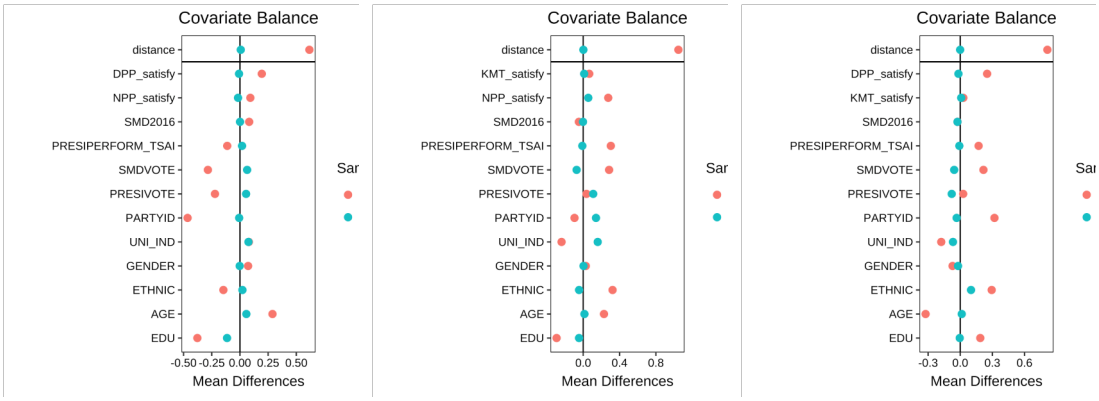




2020

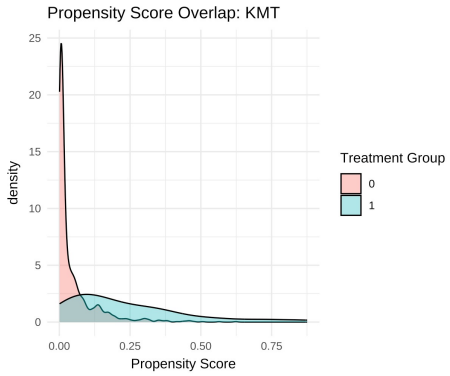
KMT Sat. DPP Sat. NPP Sat.

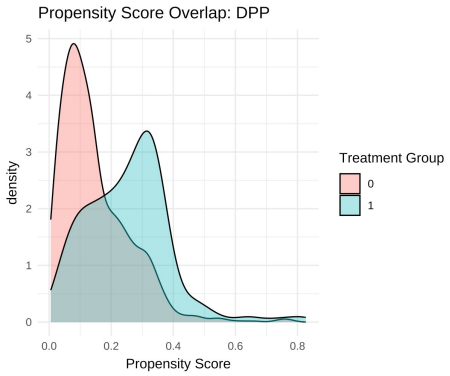


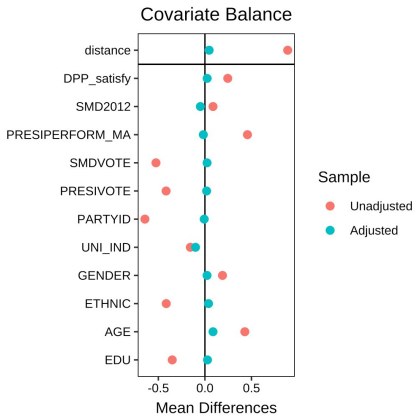


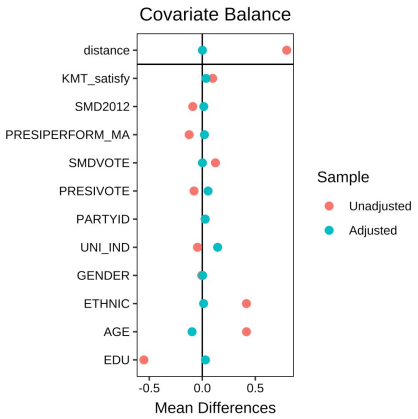
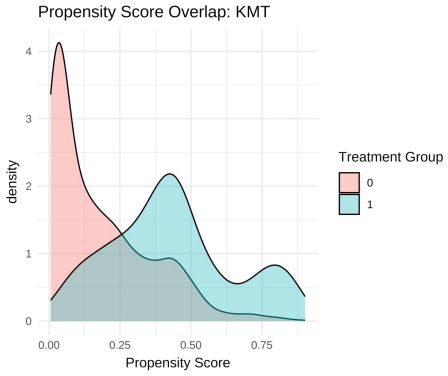
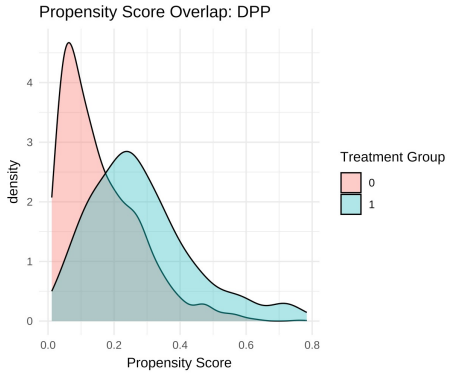
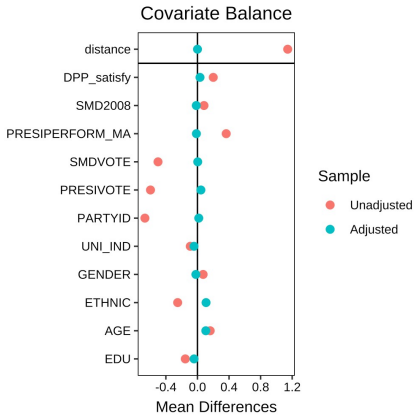
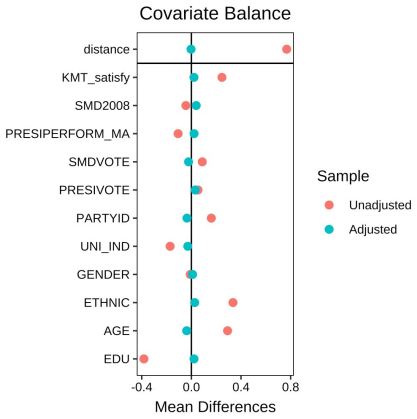
2016

KMT Sat. DPP Sat.









2012	
KMT Sat.	DPP Sat.